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Site-Selective Direct γ-Difunctionalization of Diazoenals: Application to the Synthesis of Enal-Functionalized Allenes and Furans

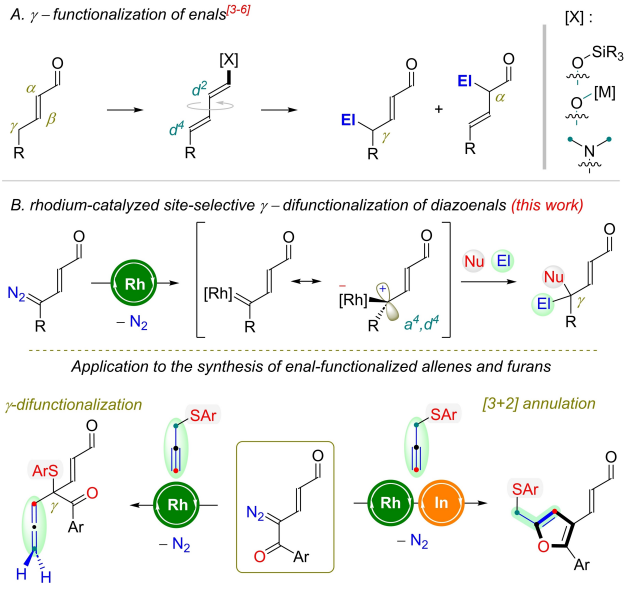
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A new approach for the synthetically important γ-functionalized enals has been developed. The strategy involves rhodium-catalyzed direct C–C & C–S bond forming site-selective γ-difunctionalization of diazoenals with aryl propargyl sulfides *via* sulfur ylide [2,3] sigmatropic rearrangement, resulting in the highly functionalized γ-allenyl(sulfanyl)enals in excellent yield at ambient temperature. This highly versatile approach constitutes a viable alternative to the remote carbonyl-directed γ-functionalization of unmodified enals which suffer from competitive side reactions. The synthetic utility of the γ-allenyl(sulfanyl)enals was demonstrated by the $InCl_3$-catalyzed cycloisomerization to the trisubstituted furanyl-enals *via* an unusual 1,4-sulfanyl migration. In addition, an operationally simple and efficient one-pot [3+2] annulation of diazoenals and propargyl sulfides, involving combined γ-difunctionalization and cycloisomerization was successfully developed for the diverse enal-functionalized furans.

Introduction

Enals are important building blocks in chemical synthesis and continue to be vital substrates for the development of organocatalysis.[1–2] However, despite the enormous synthetic utility, very few catalytic methods are available for the γ-functionalization of enals. The direct site-selective γ-functionalization of unmodified enals has been challenging.[3–7] So far, the known approaches relies on the remote carbonyl-directed γ-functionalization *via* the electron-rich dienyl motifs such as Mukaiyama dienolsilylethers,[3] metal-dienolates[4] and dienamines (Scheme 1A).[5,6] However, these approaches often suffer from the competitive α-functionalization and cycloaddition pathways of the conformationally labile 4-carbon dienyl motif.[5,8,9] Therefore, given the limitations associated with the existing strategies, the development of general, efficient, and operationally simple new catalytic approaches to the synthetically useful γ-functionalized enals would be highly valuable.

Metal carbenes derived from diazo compounds are valuable reactive intermediates that are central to a wide range of carbene transfer reactions such as σ-bond insertions (C–H, C–X, and X–H; X = heteroatom), cycloadditions, and ylide rearrangement reactions.[10,11] In particular, the ability of carbenoids to offer geminal difunctionalization through acceptor/donor reactivity provides rapid molecular complexity. Inspired by the synthetic potential of diazo compounds, we recently designed a new class of enal-functionalized diazo compounds (diazoenal).[12] We envisioned that the conformationally rigid *(E)-s-trans*-enalcarbenoid derived from diazoenal would allow direct γ-functionalization through various carbene transfer reactions, leading to the new classes of functionalized enals (Scheme 1B). It is noteworthy that, unlike dienolates or dienamines which possess d^4 reactivity, the enalcarbenoid provides dual a^4 and d^4 reactivity at the γ-carbon center, thus providing access to the unique γ-difunctionalization of enals which are not possible by the existing strategies.[3–7] Herein we report a highly efficient construction of γ-quaternary stereogenic γ-



Scheme 1. γ-Functionalization of enals and diazoenals.

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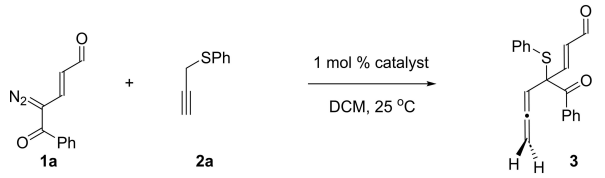
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allenyl(sulfanyl)enals by the rhodium-catalyzed direct C–C and C–S bond forming γ-difunctionalization of diazoenals with propargyl sulfides (Scheme 1B). To the best of our knowledge, this is the first report on the direct γ-difunctionalization of enals, which does not involve remote carbonyl group assistance. The synthetic utility of the γ-allenyl enals was demonstrated by the combined Rh(II), and In(III)-catalyzed [3 + 2] annulation of diazoenals and propargyl sulfides leading to the valuable enal-functionalized trisubstituted furans (Scheme 1B).

Results and Discussion

In continuation of our ongoing studies on the synthetic applications of diazoenals,[12–15] we recently reported the site-selective γ-amination *via* enalcarbenoid NH-insertion reaction with propargyl amine. The resulting donor-acceptor dienamine intermediates were successfully utilized in the construction of valuable enal-functionalized 1,4-oxazine heterocycles.[13] Inspired by these studies, we next examined the carbene transfer reaction of diazoenal **1a** and phenyl propargyl sulfide **2a** anticipating C–S & C–C bond forming direct γ-difunctionalization through the sulfur ylide [2,3] sigmatropic rearrangement.[16] Gratifyingly, a variety of dirhodium carboxylates such as $Rh_2(OAc)_4$, $Rh_2(oct)_4$, $Rh_2(tfa)_4$ and $Rh_2(esp)_2$ efficiently catalyzed the reaction producing γ-quaternized γ-allenyl(sulfanyl)enal **3** in 85–95% yield (Table 1, entries 1–4). The bulky chiral Rh_2{(*S*)-DOSP}$_4$ required longer reaction time and gave allene **3** in reduced yield (65%) with the marginal enantioselectivity (5% ee). The best yield was obtained with 1 mol % $Rh_2(OAc)_4$ at ambient conditions (95%, entry 1). The reaction was compatible with other halogenated solvents, and toluene, whereas oxygenated solvents are not compatible (See SI for full optimization table).

Table 1. Optimization of the γ-difunctionalization reaction.[a]

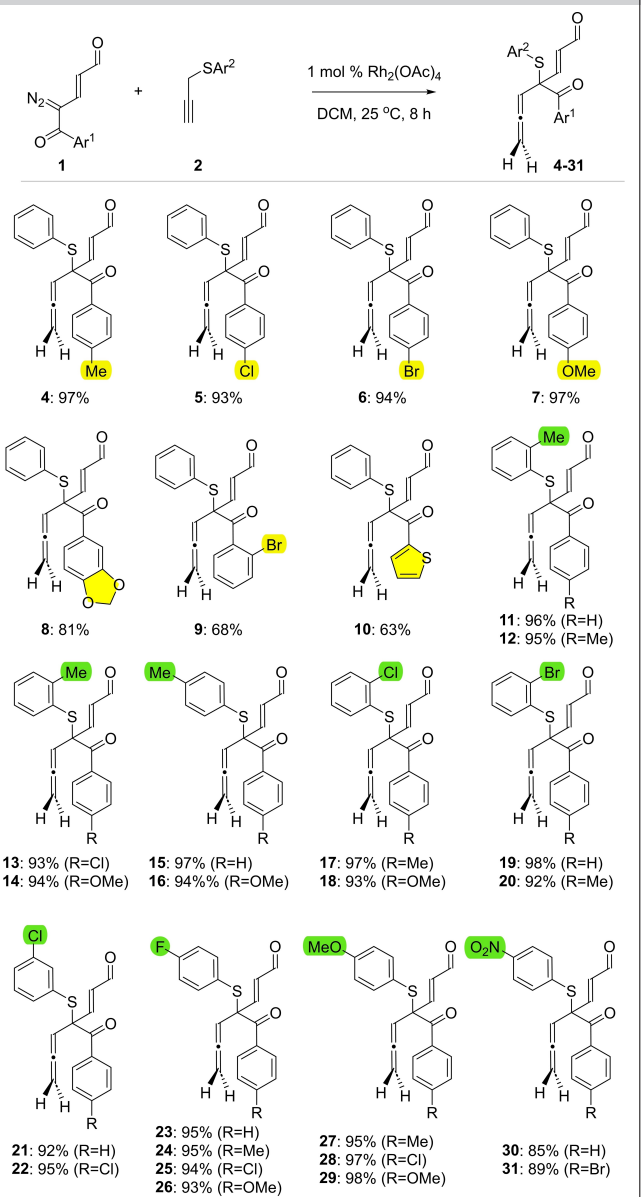


Entry	catalyst	time (h)	Yield(%)[b]
1	**$Rh_2(OAc)_4$**	**8**	**95**
2	$Rh_2(oct)_4$	8	90
3	$Rh_2(tfa)_4$	8	85
4	$Rh_2(esp)_2$	8	88
5[c]	Rh_2{(*S*)-DOSP}$_4$	18	65

[a] Reaction conditions: A solution of **1a** (0.15 mmol) in 2 mL solvent was added over 4 h to a solution of **2a** (0.15 mmol), and catalyst in 2 mL solvent, and continued the reaction until **1a** was consumed. [b] Yield of isolated product. [c] 5% ee of **3**. tfa = trifluoroacetate, esp = α,α,α′,α′-tetramethyl-1,3-benzenedipropionic acid, DOSP = (*N*-dodecylbenzenesulfonyl)prolinate.

With the optimized conditions (Table 1, entry 1) the scope of the γ-difunctionalization was examined with various diazoenals and aryl propargyl sulfides (Table 2). Though the electronic nature of the aryl substituent of diazoenal does not have much effect on the reaction (**3**–**8**: 81–97%), sterically hindered 2-bromoaryl and thiophene groups hampered the yields (**9**–**10**: 63–68%). Among the aryl propargyl sulfides, a variety of electronically distinct aryl substituents such as alkyl-, halo- (F, Cl, Br), methoxy-, and nitro- groups were well tolerated to give excellent yields of the corresponding allenyl-enals (**11**–**31**: 85–98%). However, the marginal influence of the electronic nature of the aryl group was observed. The strongly electron-withdrawing 4-NO_2 substituent gave a slightly diminished yield (**30**–**31**: 85–89%) compared to the electron-donating 4-OMe

Table 2. Substrate Scope of the γ-difunctionalization reaction.[a]



[a] Reaction was performed using optimized conditions of Table 1.

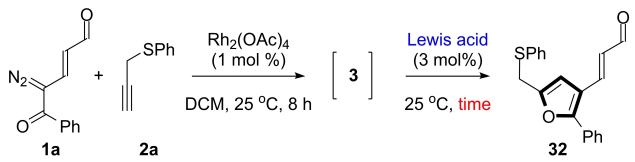
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substituent (**27**–**29**: 95–98%). Interestingly, the steric nature of the aryl group does not influence the γ-difunctionalization as the ortho-substituted aryl groups also gave excellent yields of allenes (**11**–**14** and **17**–**20**: 92–98%).

Allenes are important structural motifs present in many natural products, pharmaceuticals, and materials.[17] Moreover, allenes are valuable building blocks in organic synthesis.[18] Heteroatom functionalized allenes undergo a variety of metal-catalyzed cycloisomerizations to give heterocycles.[19] Having successfully established an efficient strategy for the γ-allenyl(sulfanyl)enals, we investigated their Lewis acid-catalyzed cycloisomerization (Table 3). Gratifyingly, exposure of γ-allenyl(sulfanyl)enal **3** to the catalytic amount of $InCl_3$ (3 mol %) in DCM smoothly furnished the trisubstituted furanyl-enal **32** in quantitative yield *via* an unusual 1,4-sulfanyl migration.[20] At this stage, we investigated the feasibility of a one-pot γ-difunctionalization and cycloisomerization reactions through Rh(II)/In(III) relay catalysis. To our delight, slow addition of diazoenal **1a** to phenyl propargyl sulfide **2a**, $Rh_2(OAc)_4$ (1 mol %), and $InCl_3$ (3 mol %) in DCM at ambient temperature produced the furan **32** in 76% yield through an overall formal [3+2] annulation. Further control experiments revealed that addition of $InCl_3$ catalyst after completion of γ-difunctionalizaton provides an excellent yield of furan (94%, Table 3, entry 1). Among the other In(III)-catalysts, $InBr_3$ also gave high yield (entry 2) whereas $In(OTf)_3$ led to the decomposition of allene intermediate (entry 3). Other Lewis acid catalysts such as AuCl, $AuCl_3$, $PtCl_2$, $Sc(OTf)_3$, $Zn(OTf)_2$ are not suitable for the cyclo-

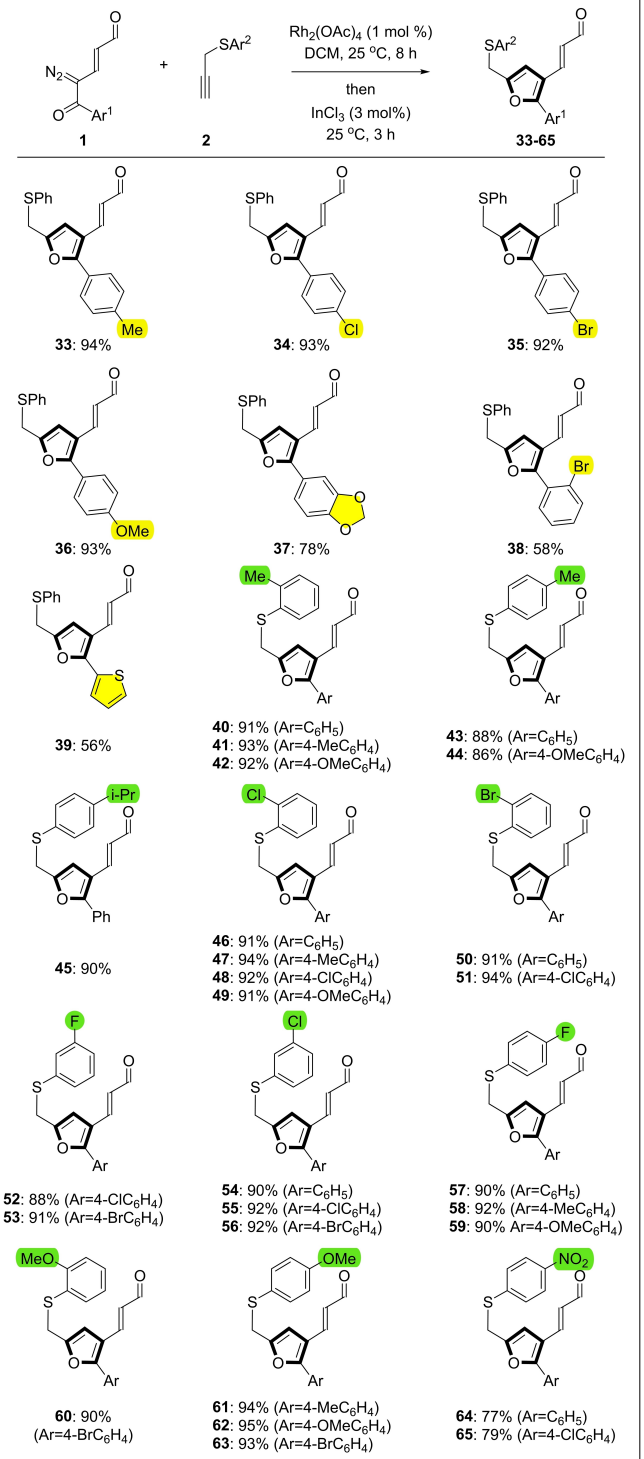
Table 3. Optimization of the [3+2] annulations.[a]



Entry	Lewis acid	Time (h)	Yield (%)[b]
1	**$InCl_3$**	**3**	**94**
2	$InBr_3$	3	85
3[c]	$In(OTf)_3$	3	0
4[c]	AuCl	3	0
5[c]	$AuCl_3$	3	0
6[c]	$PtCl_2$	3	0
7[c]	$Sc(OTf)_3$	3	0
8[d]	$Zn(OTf)_2$	8	0
9	$ZnCl_2$	8	82
10[d]	none	10	0
11[e]	$InCl_3$	2	75

[a] Reaction conditions: A solution of **1a** (0.23 mmol) in DCM (2 mL) was added over 4 h to a solution of **2a** (0.15 mmol), rhodium catalyst in DCM (2 mL) and continued the reaction until the consumption of **2a** (8 h), then Lewis acid catalyst was added to the reaction mixture and continued the reaction until the specified time. [b] Yield of isolated product. [c] Allene intermediate was decomposed. [d] Allene intermediate was unreacted. [e] Reaction was performed at 40 °C

Table 4. Substrate Scope of the [3+2] Annulation Reaction.[a]



[a] Reaction was performed using optimized conditions of Table 3.

isomerization reaction (entries 4–8). Interestingly, $ZnCl_2$ also effective despite longer reaction time (82%, entry 9). Furan was not obtained without the Lewis acid catalyst (entry 10). Increasing the reaction temperature gave reduced yield (entry 11). Thus, an optimum yield of 94% was obtained with the

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1 mol% $Rh_2(OAc)_4$ and 3 mol% $InCl_3$ at the ambient temperature (entry 1).

The scope of the [3+2] annulation involving γ-difunctionalization/cycloisomerization was investigated with diverse enaldiazo ketones **1** and aryl propargyl sulfides **2** (Table 4). Excellent yields of furans **33**–**37** (87–94%) were obtained with alkyl, halo, and oxygenated aryl enaldiazo ketones indicating that reaction is not sensitive to the electronic nature of the aryl ring. However, the sterically hindered 2-Br aryl and heteroaryl thiophene substituted diazoenal hampered the reaction (**38**–**39**, 56–58%). In case of aryl propargyl sulfides, diverse alkyl-, halo-, and methoxy-substitutions on the aryl ring are well tolerated leading to the furans **40**–**63** in excellent yield (86–96%). Whereas electron-withdrawing nitro-group resulted in slightly diminished yield of furans **64**–**65** (77–79%). With respect to the steric influence, the 2-methyl, 2-halo-, and 2-methoxy aryl sulfides also smoothly participated in the reaction (**40**–**42**, **46**–**51**, and **60**).

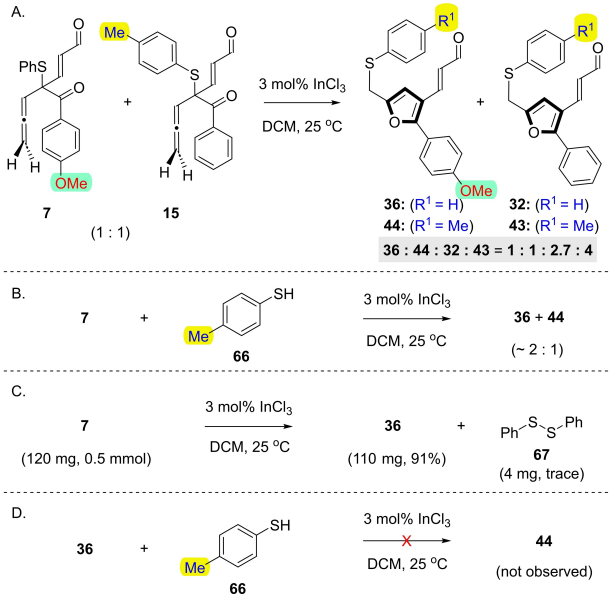
To understand the mechanism of the furan formation, a series of control experiments were carried out (Scheme 2). A 1:1 mixture of allene **7** and **15** was subjected to the rearrangement reaction in the presence of 3 mol % $InCl_3$. Interestingly, in addition to the anticipated furans **36** and **43**, the reaction also produced considerable amounts of crossed furan products **44** and **32** through intermolecular sulfide migration (Scheme 2A). Performing the cycloisomerization of allene **7** in the presence of 4-methylthiophenol **66** produced furan **36** along with the crossed furan **44** in about a 2:1 ratio (Scheme 2B). Further, a large-scale cycloisomerization reaction of **7** led to the detection of a trace amount of diphenyl disulfide **67** (Scheme 2C). However, treating furan **36** with 4-methylthiophenol **66** in the presence of $InCl_3$ failed to produce the crossed furan **44** (Scheme 2D), indicating that sulfanyl migration is irreversible.

Based on the control experiments, a plausible mechanism for the [3+2] annulation *via* γ-difunctionalization and cycloisomerization reaction, involving Rh(II) and In(III) catalytic cycles is depicted in Scheme 3. The Rh-enalcarbenoid **68** derived from diazoenal **1** reacts with propargyl sulfide **2** to give sulfonium ylide **69A**. Subsequent [2,3] sigmatropic rearrangement of **69A** or the metal-free ylide **69B** leads to the γ-allenyl(sulfanyl)enal **70**. The low enantioselectivity (<5%) observed for Table 1, entry 5 may support the metal free ylide rearrangement. In the In-catalysis, intramolecular allylic sulfide displacement *via* 5-*endo-dig* heterocyclization of In(III) activated allenyl sulfide **71** results in the unstable 2-methylene-2*H*-furan-1-ium and Indium-thiolate ion pair **72**. The involvement of this ion pair intermediate supports the results of the control experiments (Scheme 2). Finally, the recombination of ion pair **72** by the thiolate nucleophilic addition completes the 1,4-sulfanyl migration[21] and furnishes the enal-functionalized furan product **73**. Notably, the overall annulation reaction involves only N_2 as the by-product.

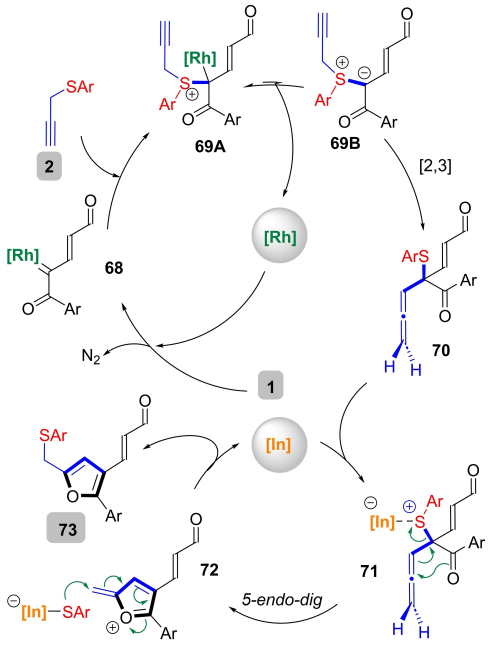
The synthetic utility of the furanyl-enal **35** was demonstrated by further transformations (Scheme 4). Chemoselective oxidation using *m*-CPBA, gave a mixture of sulfone **74** and sulfoxide **75** in 76% and 22% yield respectively.[22]. The structure of **74** has been confirmed by the single crystal X-ray diffraction. Olefination of furanyl-enal **35** with phosphorane **76** produced the valuable dienone-tethered furan **77** in 81% yield.

Conclusions

In summary, we have developed a new approach for the synthetically important γ-difunctionalized enals. The strategy involves rhodium-catalyzed direct C–C & C–S bond forming site-selective γ-difunctionalization of diazoenals with aryl propargyl sulfides *via* sulfur ylide [2,3] sigmatropic rearrangement



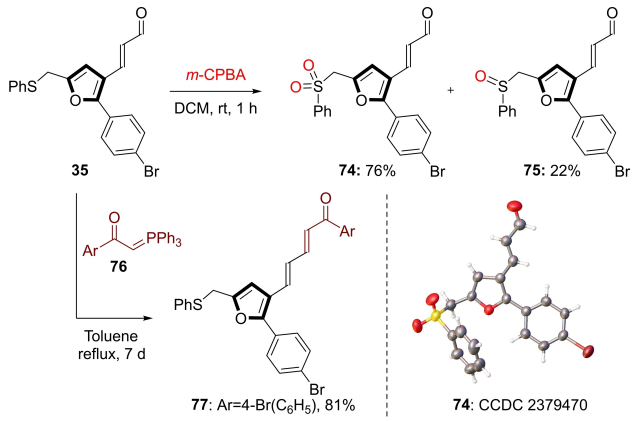
Scheme 2. Control experiments.



Scheme 3. A plausible mechanism of the γ-difunctionalization and cycloisomerization.

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Scheme 4. Synthetic applications of furanyl-enals.

and provides functionally rich γ-allenyl(sulfanyl)enals in excellent yield at ambient temperature. The synthetic utility of the new enals was demonstrated by the $InCl_3$-catalyzed cycloisomerization to the trisubstituted furanyl-enals *via* an unusual 1,4-sulfanyl migration. Moreover, an operationally simple and efficient one-opt [3 + 2] annulation of diazoenals and aryl propargyl sulfides gave diverse enal-functionalized furans. Further study on the synthetic applications of γ-difinctionalization and [3 + 2] annulation is ongoing.

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Conflict of Interests

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords: Allene · Carbenoid · Enal · Diazoenal · Furan

[1] a) A. Berkessel, H. Gröger, *Asymmetric Organocatalysis From Biomimetic Concepts to Applications in Asymmetric Synthesis*. Wiley, **2005**, New York;; b) A. J. B. Watson, D. W. C. MacMillan, *Enantioselective organocatalysis involving iminium, enamine, SOMO, and photoredox activation*. Catalytic Asymmetric Synthesis (3rd Edn), Ed. Ojima, I. **2010**, 39–57;; c) B. List, Asymmetric Organocatalysis *Top. Curr. Chem*, Vol. *291*, Springer **2010**; d) D. W. C. MacMillan, A. J. B. Watson, in *Science of Synthesis; B. List*, Ed.; Thieme: Stuttgart, Germany, **2012**; e) A. T. Biju, *N-Heterocyclic Carbenes in Organocatalysis*; Wiley-VCH Verlag GmbH & Co. KGaA, **2019**.
[2] a) G. Lelais, D. W. C. MacMillan, *Aldrichimica Acta* **2006**, *39*, 79–87; b) B. List, *Chem. Commun.* **2006**, *8*, 819–824; c) A. Erkkilä, I. Majander, P. M. Pihko, *Chem. Rev.* **2007**, *107*, 5416–5470; d) D. Enders, O. Niemeier, A. Henseler, *Chem. Rev.* **2007**, *107*, 5606–5655; e) D. W. C. MacMillan, *Nature* **2008**, *455*, 304–308; f) P. Melchiorre, M. Marigo, A. Carlone, G. Bartoli, *Angew. Chem. Int. Ed.* **2008**, *47*, 6138–6171; g) V. Nair, S. Vellalath, B. P. Babu, *Chem. Soc. Rev.* **2008**, *37*, 2691–2698; h) D. M. Flanigan, F. Romanov-Michailidis, N. A. White, T. Rovis, *Chem. Rev.* **2015**, *115*, 9307–9387; i) B. S. Donslund, T. K. Johansen, P. H. Poulsen, K. S. Halskov, K. A. Jørgensen, *Angew. Chem. Int. Ed.* **2015**, *54*, 13860–13874.
[3] a) T. Mukaiyama, A. Ishida, *Chem. Lett.* **1975**, *4*, 319322; b) G. Casiraghi, F. Zanardi, G. Appendino, G. Rassu, *Chem. Rev.* **2000**, *100*, 19291972; c) S. E. Denmark, J. R. Heemstra, G. L. Beutner, *Angew. Chem. Int. Ed.* **2005**, *44*, 4682–4698.
[4] H.-J. Zhang, A. W. Schuppe, S.-T. Pan, J.-X. Chen, B.-R. Wang, T. R. Newhouse, L. Yin, *J. Am. Chem. Soc.* **2018**, *140*, 5300–5310.
[5] a) D. B. Ramachary, Y. V. Reddy, *Eur. J. Org. Chem.* **2012**, 865–887; b) H. Jiang, Ł. Albrecht, K. A. Jørgensen, *Chem. Sci.* **2013**, *4*, 2287–2300; c) I. D. Jurberg, I. Chatterjee, R. Tannert, P. Melchiorre, *Chem. Commun.* **2013**, *49*, 4869–4883; d) C. Schneider, F. Abels, *Org. Biomol. Chem.* **2014**, *12*, 3531–3543; e) V. Marcos, J. Alemán, *Chem. Soc. Rev.* **2016**, *45*, 6812–6832; f) J. L. Vicario, *Synlett* **2016**, *27*, 1006–1021; g) L. Klier, F. Tur, P. H. Poulsen, K. A. Jørgensen, *Chem. Soc. Rev.* **2017**, *46*, 1080–1102.
[6] a) M. Silvi, E. Arceo, I. D. Jurberg, C. Cassani, P. Melchiorre, *J. Am. Chem. Soc.* **2015**, *137*, 6120–6123; b) L. Naesborg, V. Corti, L. A. Leth, P. H. Poulsen, K. A. Jørgensen, *Angew. Chem. Int. Ed.* **2018**, *130*, 1622–1626.
[7] J. Mo, X. Chen, Y. R. Chi, *J. Am. Chem. Soc.* **2012**, *134*, 8810–8813.
[8] a) K. Yachi, H. Shinokubo, K. Oshima, *J. Am. Chem. Soc.* **1999**, *121*, 9465–9466; b) S.-H. Chen, B.-C. Hong, C. F. Su, S. Sarshar, *Tetrahedron Lett.* **2005**, *46*, 8899–8903; c) N. Utsumi, H. Zhang, F. Tanaka, C. F. Barbas III, *Angew. Chem. Int. Ed.* **2007**, *46*, 1878–1880; d) A. S. González, R. G. Arrayás, M. R. Rivero, J. C. Carretero, *Org. Lett.* **2008**, *10*, 4335–4337.
[9] a) R. M. de Figueiredo, R. Fröhlich, M. Christmann, *Angew. Chem. Int. Ed.* **2008**, *47*, 1450–1453; b) B. Han, Z.-Q. He, J.-L. Li, R. Li, K. Jiang, T.-Y. Liu, Y.-C. Chen, *Angew. Chem. Int. Ed.* **2009**, *48*, 5474–5477; c) G. Bergonzini, S. Vera, P. Melchiorre, *Angew. Chem. Int. Ed.* **2010**, *49*, 9685–9688.
[10] a) M. P. Doyle, M. A. McKervey, T. Ye, *Modern Catalytic Methods for Organic Synthesis with Diazo Compounds: From Cyclopropanes to Ylides*; Wiley: New York, **1998**; b) Modern Rhodium-Catalyzed Organic Reactions (Ed: P. A. Evans), Wiley-VCH, Weinheim **2005**.
[11] a) M. P. Doyle, *Chem. Rev.* **1986**, *86*, 919–939; b) M. P. Doyle, D. C. Forbes, *Chem. Rev.* **1998**, *98*, 911–936; c) H. Lebel, J.-F. Marcoux, C. Molinaro, A. B. Charette, *Chem. Rev.* **2003**, *103*, 977–1050; d) H. M. L. Davies, J. R. Manning, *Nature* **2008**, *451*, 417–424; e) Z. Zhang, J. Wang, *Tetrahedron* **2008**, *64*, 6577–6605; f) A. Padwa, *Chem. Soc. Rev.* **2009**, *38*, 3072–3081; g) S.-F. Zhu, Q.-L. Zhou, *Acc. Chem. Res.* **2012**, *45*, 1365–1377; h) X. Guo, W. Hu, *Acc. Chem. Res.* **2013**, *46*, 2427–2440; i) A. Ford, H. Miel, A. Ring, C. N. Slattery, A. R. Maguire, M. A. McKervey, *Chem. Rev.* **2015**, *115*, 9981–10080.
[12] a) S. G. Dawande, V. Kanchupalli, J. Kalepu, H. Chennamsetti, B. S. Lad, S. Katukojvala, *Angew. Chem. Int. Ed.* **2014**, *53*, 7831–7835; b) S. G. Dawande, V. Kanchupalli, B. S. Lad, J. Rai, S. Katukojvala, *Org. Lett.* **2014**, *16*, 3700–3703.
[13] J. Kalepu, S. Katukojvala, *Angew. Chem. Int. Ed.* **2016**, *55*, 7831–7835.
[14] a) V. Kanchupalli, S. Katukojvala, *Angew. Chem. Int. Ed.* **2018**, *57*, 5433–5437; b) P. K. Mandal, S. Katukojvala, *Chem. Eur. J.* **2024**, *30*, e202303862; c) P. K. Mandal, R. Chakrawarti, S. Katukojvala, *Org. Lett.* **2024**, *26*, 3463–3467; d) H. Chennamsetti, K. S. Rathore, S. Chatterjee, P. K. Mandal, S. Katukojvala, *JACS Au* **2024**, *4*, 2099–2107; e) P. K. Mandal, S. Patel, S. Katukojvala, *Org. Biomol. Chem.* **2024**, *22*, 5734–5738.
[15] J.-Q. Wu, Z. Yang, S.-S. Zhang, C.-Y. Jiang, Q. Li, Z.-S. Huang, H. Wang, *ACS Catal.* **2015**, *5*, 6453–6457.
[16] a) Y. Zhang, J. Wang, *Coord. Chem. Rev.* **2010**, *254*, 941–953; b) Z. Sheng, Z. Zhang, C. Chu, Y. Zhang, J. Wang, *Tetrahedron* **2017**, *73*, 4011–4022; c) A. C. S. Reddy, P. Anbarasan, *Org. Lett.* **2019**, *21*, 9965–9969; d) J. Yang, J. Wang, H. Huang, G. Qin, Y. Jiang, T. Xiao, *Org. Lett.* **2019**, *21*, 2654–2657; e) F. He, S. Jana, R. M. Koenings, *J. Org. Chem.* **2020**, *85*, 11882–11891; f) V. Kanchupalli, L. A. Thorbole, J. Kalepu, D. Joseph, M. Arshad, S. Katukojvala, *Org. Lett.* **2022**, *24*, 3850–3854; g) Z. Fang, Y. Wang, Y. Ma, X. Han, Z. Liu, Y. Ning, *J. Org. Chem.* **2023**, *88*, 9677–9685;

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1861471x, 2025, 1, Downloaded from https://aces.onlinelibrary.wiley.com/doi/10.1002/asia.202401081 by Fudan University, Wiley Online Library on [25/02/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

h) Y.-D. Hao, J. Liang, Z.-Q. Lin, T.-L. Huang, Y.-D. Xu, L. Guo, Z.-Z. Yang, Y. Wu, *Org. Chem. Front.* **2023**, *10*, 4038–4042; i) Y.-D. Hao, Z.-Q. Lin, X.-Y. Guo, J. Liang, C.-K. Luo, Q.-T. Wang, L. Guo, Y. Wu, *Chin. Chem. Lett.* **2024**, *35*, 108834; j) Y. Chen, G. Lv, Z. Li, M. Ye, T. Huang, L. Hai, Y. Wu, *Green Synth. Catal.* **2024**, *5*, 77–79.

[17] a) A. Hoffmann-Röder, N. Krause, *Angew. Chem. Int. Ed.* **2004**, *43*, 1196–1216; b) P. Rivera-Fuentes, F. Diederich, *Angew. Chem. Int. Ed.* **2012**, *51*, 2818–2828.

[18] a) Modern Allene Chemistry (Eds.: N. Krause, A. S. K. Hashmi), Wiley-VCH, Weinheim, **2004**; b) S. Yu, S. Ma, *Angew. Chem. Int. Ed.* **2012**, *51*, 3074–3112; c) B. S. Schreib, M. Son, F. A. Aouane, M.-H. Baik, E. M. Carreira, *J. Am. Chem. Soc.* **2021**, *143*, 21705–21712.

[19] a) A. S. K. Hashmi, in *Modern Allene Chemistry, Vol. 2* (Eds.:), Wiley-VCH, Weinheim, **2004**, 877–923; b) N. Krause, C. Winter, *Chem. Rev.* **2011**, *111*, 1994–2009; c) A. V. Gulevich, A. S. Dudnik, N. Chernyak, V. Gevorgyan, *Chem. Rev.* **2013**, *113*, 3084–3213.

[20] a) M. Bao, Y. Qian, H. Su, B. Wu, L. Qiu, W. Hu, X. Xu, *Org. Lett.* **2018**, *20*, 5332–5335; b) B. S. Lad, S. Katukojvala, *ACS Catal.* **2018**, *8*, 11807–11814; c) S. Zhu, F. Li, C. Empel, S. Jana, C. Pei, R. M. Koenigs, *Adv. Synth. Catal.* **2022**, *364*, 3149–3154; d) C. Hong, S. Yu, Z. Liu, Z. Xu, Y. Zhang, *J. Org. Chem.* **2022**, *87*, 11979–11988; e) X. Lei, J. Feng, Q. Guo, Y. Li, J. Shi, *Org. Lett.* **2023**, *25*, 7338–7343.

[21] L. Peng, X. Zhang, M. Ma, J. Wang, *Angew. Chem. Int. Ed.* **2007**, *46*, 1905–1908.

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